

HENRY GUO

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EDUCATION

University of Texas at Austin

Bachelor of Science in Electrical and Computer: GPA: 3.91

Honors: University Honors (Fall 2024 - Summer 2025)

Expected Graduation 2028

EXPERIENCE

Teaching Assistant for Circuit Theory - ECE 411

Fall 2025

- Led technical labs using oscilloscopes, waveform generators, breadboards, and power supplies.
- Assisted over 40 students with measuring waveforms, debugging circuits, and interpreting circuit behavior
- Hosted office hours teaching topics like AC/DC analysis, phasors, complex impedance, and power calculations

Verilog Hex Counter and Digital Logic Labs- ECE 316

Fall 2025

- Designed Verilog modules in Xilinx Vivado for a rising-edge detector, hexadecimal counter, seven-segment display sequencer, ripple-carry adder, and 4-bit carry-lookahead adder.
- Developed Verilog testbenches and simulations for testing modules, clocks, reset behavior, and timers.
- Verified hardware output and debugged waveforms with the Artix-7 FPGA board

Microcontroller Game Development - Introduction to Embedded Systems

Spring 2025

- Developed and implemented an arcade game on a custom PCB using the MSPM0+ Microcontroller
- Interfaced 4 buttons, slide pots, LEDs, LCD display, and custom DAC
- Programmed game logic in C while designing with KiCad, including soldering and debugging

Project Manager Assistant

January - July 2024

Musical Arts Center of San Antonio Inc

- Optimized teachers' classroom schedules with Excel and Mindbody and minimized conflicts for all 35 instructors
- Renovated a student achievement board with Canva, showcasing awards of over 100 students
- Maintained consistent, professional correspondence with students and teachers

SKILLS

- Relevant Coursework: Digital Logic Design, Circuit Theory, Differential Equations with Linear Algebra, Software Design and Implementation, Signals and Systems
- Technical Skills: Verilog HDL, waveform debugging, C, Python, ARM assembly, KiCad, Microsoft Excel, PowerPoint, LTspice, oscilloscopes, waveform generators, multimeters, breadboarding, soldering, Linux, BIOS/UEFI
- Language Fluency: Native in English, Conversational in Mandarin

PROJECTS

Home Electronics Repair - Self project

2024 - Present

- Practiced safe high-voltage practice including discharge checks, isolation, and careful multimeter measurements.
- Desoldered and replaced damaged through-hole and surface-mount components using soldering tools, flux, solder wick, and continuity testing.

Linux Systems and PC Building - Self project

2024 - Present

- Assembled desktop systems from components including CPU, cooler, motherboard, RAM, GPU, storage drives, PSU, case fans, and front-panel connectors.
- Configured and maintained desktop systems using BIOS/UEFI setup, boot-order configuration, storage partitioning, and OS installation.